

Replication Workflow: Generative AI-Driven Accelerated Discovery of Passivation Molecules for Perovskite Solar Cells

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Abstract

We describe our workflow for replicating the computational pipeline of Fajar et al. (2026), which combines a SMILES-X binary classifier with iterative GPT-2 fine-tuning to discover novel passivation molecules for perovskite solar cells. Our replication uses the original authors’ dataset (314 labeled molecules), reproduces the three-stage pipeline (classification, generation, filtering), and distributes the computationally intensive generation phase across multiple NVIDIA DGX Spark systems. We report preliminary results, discuss reproducibility challenges encountered, and document deviations from the published method.

1 Introduction

Fajar et al. [1] proposed a generative AI framework for accelerated discovery of passivation molecules for perovskite solar cells (PSCs). Their approach combines discriminative and generative models in an iterative loop: a SMILES-X classifier [2] identifies promising molecules, while a fine-tuned GPT-2 model generates novel candidates. Over three iterative cycles, the pipeline produces over 100,000 chemically unique, novel (CUN) molecules, of which more than 80% are predicted effective passivators. After physicochemical filtering and clustering, three candidates were experimentally validated, with one—4-methylbenzamide (MBA)—achieving a 22.05% power conversion efficiency in inverted PSC devices.

Our goal is to independently replicate this computational pipeline from the published description and supplementary data. This document describes our workflow, infrastructure, preliminary results, and challenges encountered.

2 Original Pipeline Overview

The Fajar et al. framework consists of three stages:

2.1 Stage 1: Discriminative Model

A dataset of 314 experimentally reported passivation molecules is assembled from the literature, each labeled with a normalized PCE improvement ($\Delta\text{PCE}_{\text{norm}}$). Molecules with $\Delta\text{PCE}_{\text{norm}} \geq 0.10$ are classified as effective (class 1); those below as ineffective (class 0). A SMILES-X binary classifier—a BiLSTM network with soft attention operating on tokenized SMILES strings—is trained using 5-fold cross-validation.

The dataset is then augmented by querying PubChem for molecules with $\geq 80\%$ Tanimoto similarity (based on Morgan fingerprints) to the top-performing class 1 molecules ($\Delta\text{PCE}_{\text{norm}} > 0.16$). This yields approximately 15,500 additional molecules. All augmented molecules are

classified; those predicted as class 1 are combined with the original class 1 molecules to form the training set T_1 ($\sim 11,000$ molecules).

A critical reproducibility detail: the classification threshold was optimized at 0.47 rather than the default 0.5, maximizing F1 score on the validation set.

2.2 Stage 2: Generative Model (Iterative)

A GPT-2 language model is fine-tuned on the SMILES representations of molecules in T_1 . SMILES augmentation (random atom re-ordering) is applied to increase training diversity. The fine-tuned model then generates 100,000 novel molecules per cycle via autoregressive sampling at temperature $T = 0.9$.

Generated molecules are classified by the Stage 1 model. Those predicted as class 1 are added to the training set for the next cycle. This iterative process runs for exactly three cycles:

$$T_2 = T_1 \cup \{\text{class-1 molecules from } G_1\} \quad (1)$$

$$T_3 = T_2 \cup \{\text{class-1 molecules from } G_2\} \quad (2)$$

$$\text{CUN} = \text{class-1}(G_1) \cup \text{class-1}(G_2) \cup \text{class-1}(G_3) \quad (3)$$

The authors specifically limited to three cycles to avoid model collapse—a phenomenon where iterative self-training on synthetic data leads to mode collapse and loss of diversity.

2.3 Stage 3: Physicochemical Filtering

The CUN molecules are subjected to seven physicochemical filters:

1. Synthetic accessibility (SA) score ≤ 6
2. No PAINS (Pan-Assay Interference Compounds) substructures
3. Hydrogen bond donors (HBD): 0–2
4. Hydrogen bond acceptors (HBA): 2–5
5. Topological polar surface area (TPSA): 50–120 Å²
6. HOMO-LUMO energy gap: 1.5–5.0 eV
7. Dipole moment: 1.5–4.0 D

Filters 1–5 are computed using RDKit. Filters 6–7 require semi-empirical quantum chemistry (xTB) or DFT calculations. After filtering, molecules are clustered into 10 groups via agglomerative clustering on Morgan fingerprints, and one representative is sampled from each cluster.

3 Replication Workflow

3.1 Data Acquisition

We obtained the original dataset directly from the corresponding author, Dr. Guo Zhanglin at Kyushu University, who provided the complete `dataset.xlsx` file with all 12 sheets (314 base molecules plus generated sets from each cycle). We also collected the five source papers referenced for the original 314-molecule compilation.

The original code repository is available at github.com/adroitfajar/pvmol-gen, and the SMILES-X classifier framework at github.com/Lambard-ML-Team/SMILES-X.

Table 1: Compute infrastructure for replication

Machine	Hardware	GPU	Role
spark-36ac	DGX Spark	NVIDIA GB10, 120 GB RAM	Primary training & gen
spark-95fe	DGX Spark	NVIDIA GB10, 120 GB RAM	Parallel generation
spark-9611	DGX Spark	NVIDIA GB10, 120 GB RAM	Parallel generation

3.2 Infrastructure

Our replication runs on the following compute resources, all accessed at zero cost through Argonne National Laboratory:

Software environment: Python 3.11, PyTorch 2.10, Transformers 5.5, RDKit (via conda-forge), scikit-learn 1.8, Accelerate 1.13. Environments managed with Miniforge on all three machines.

3.3 Stage 1: Classifier Training

We rebuilt the SMILES-X classifier from the published architecture description: embedding layer $\rightarrow$ bidirectional LSTM $\rightarrow$ time-distributed dense layer $\rightarrow$ soft attention $\rightarrow$ sigmoid output. Hyperparameters follow the original SMILES-X defaults [2]: embedding dimension 128, LSTM units 64, dense units 64, learning rate $10^{-3.9}$.

SMILES tokenization uses the original SMILES-X regex pattern, which handles aromatic atoms, charges, ring closures, and stereochemistry tokens. Exhaustive SMILES augmentation (all valid atom orderings) is applied during training.

Table 2: Classifier performance: our replication vs. reported values

Model	F1 (ours)	AUC (ours)	F1 (paper)	AUC (paper)
SMILES-X (neural)	0.709 ± 0.028	0.743 ± 0.062	0.80	0.88
GBM baseline	0.663 ± 0.065	0.693 ± 0.103	—	—

5-Fold Cross-Validation Results. Our classifier underperforms the reported values (F1 = 0.71 vs. 0.80; AUC = 0.74 vs. 0.88). Potential causes include:

- **Architecture details:** the original SMILES-X paper specifies larger default hyperparameters (embed=512, LSTM=128, dense=128) than what we used. It is unclear which configuration Fajar et al. employed.
- **PyTorch version:** the original work likely used an earlier PyTorch version; LSTM behavior and initialization have changed across versions.
- **Augmentation strategy:** the exact SMILES enumeration method can significantly impact training set composition.

Despite the gap, we proceeded with our classifier (following the decision to investigate after obtaining end-to-end results), using the optimized threshold of 0.47 as described in the paper.

3.4 Stage 2: Iterative Generation

3.4.1 Cycle 1

GPT-2 (124M parameters) was fine-tuned on T_1 (11,086 molecules). After $5\times$ SMILES augmentation, the training set contained $\sim 11,359$ unique SMILES strings. Training used the Hugging

Face **Trainer** with early stopping (patience = 5 epochs), batch size 8, fp16 precision, AdamW with weight decay 0.01 and 500 warmup steps.

Generation was performed autoregressively: the model receives a [PAD] token as a seed and generates up to 100 tokens at temperature 0.9. Each generated string is validated and canonicalized with RDKit. The target is 100,000 chemically valid, unique, novel molecules per cycle.

Cycle 1 generated 100,000 valid molecules (G_1). Classification yielded 17,884 class-1 molecules (17.9%), with a mean predicted probability of 0.364.

3.4.2 Cycle 2

The training set T_2 was constructed by combining T_1 (11,086) with class-1 molecules from G_1 (17,884), yielding 28,968 unique molecules. GPT-2 was fine-tuned from scratch on T_2 (not continued from the Cycle 1 checkpoint, consistent with the paper’s description of training “on the expanded dataset” each cycle).

After $5\times$ augmentation: $\sim 29,491$ training SMILES. Training completed in 15 epochs (128.7 minutes on the GB10 GPU), with early stopping triggered at $\text{eval_loss} = 0.1504$.

Parallel Generation. Cycle 2 generation revealed a computational bottleneck: at ~ 32 attempts/second on a single machine with $\sim 5\%$ valid molecule yield, reaching 100,000 valid molecules would require approximately 19 hours. To address this, we distributed generation across all three DGX Spark systems (Table 1).

Each machine runs an independent generation shard with a different random seed, ensuring diversity:

- **Shard 0** (spark-36ac): Seed 42, running as part of the main pipeline
- **Shard 1** (spark-95fe): Seed 1042, target 45,000 molecules
- **Shard 2** (spark-9611): Seed 2042, target 45,000 molecules

The shards run on GPU (NVIDIA GB10, CUDA 13.0), achieving approximately 70 attempts/second per machine—a $2.2\times$ speedup over CPU. The combined generation rate across three machines is approximately 170 attempts/second, reducing the projected generation time from 19 hours to approximately 3 hours.

After all shards complete, the results are merged on the primary machine. Duplicate molecules across shards (expected to be rare given different random seeds) are removed during merging.

3.4.3 Cycle 3

Cycle 3 follows the same procedure: classify G_2 , build $T_3 = T_2 \cup \text{class-1}(G_2)$, fine-tune GPT-2 on T_3 , and generate G_3 using parallel sharding. The final CUN set is the union of all class-1 molecules from all three cycles.

3.5 Stage 3: Physicochemical Filtering

Of the seven filters, five (SA score, PAINS, HBD, HBA, TPSA) are computed directly with RDKit. The remaining two (HOMO-LUMO gap and dipole moment) require quantum chemical calculations. We use GFN2-xTB [3], a semi-empirical tight-binding method, installed via conda-forge on the DGX Spark systems.

For each candidate molecule:

1. Generate a 3D conformer using RDKit’s ETKDG method

Algorithm 1 Parallel Molecule Generation for Cycle k

Require: Trained GPT-2 model M_k , exclusion set E_k , N target molecules, S shards

```
1: target_per_shard  $\leftarrow \lceil N/S \rceil$ 
2: for each shard  $s \in \{0, 1, \dots, S-1\}$  in parallel do
3:   Set random seed to  $42 + 1000 \cdot s$ 
4:    $G_s \leftarrow \emptyset$ 
5:   while  $|G_s| < \text{target\_per\_shard}$  do
6:     Generate batch of SMILES from  $M_k$  at temperature 0.9
7:     for each generated SMILES  $x$  do
8:        $x' \leftarrow \text{Canonicalize}(x)$  using RDKit
9:       if  $x'$  is valid and  $x' \notin G_s$  and  $x' \notin E_k$  then
10:         $G_s \leftarrow G_s \cup \{x'\}$ 
11:      end if
12:    end for
13:  end while
14:  Save  $G_s$  to  $G\{k\}\_\text{shard}\{s\}.\text{csv}$ 
15: end for
16:  $G_k \leftarrow G_0 \cup G_1 \cup \dots \cup G_{S-1}$  ▷ Remove inter-shard duplicates
17: return  $G_k$ 
```

2. Optimize geometry with GFN2-xTB

3. Extract HOMO-LUMO gap and dipole moment from the xTB output

Molecules passing all seven filters are clustered into 10 groups using agglomerative clustering (Ward linkage) on 2048-bit Morgan fingerprints (radius 2). One representative is randomly sampled from each cluster.

4 Preliminary Results and Observations

4.1 Generation Yield

Table 3 shows the generation yield across cycles. The declining marginal yield within each cycle reflects the increasing difficulty of finding novel, valid molecules as the generated set grows.

Table 3: Generation statistics (Cycle 2 in progress)

Cycle	Training set	Generated	Class-1	Class-1 %
1	T_1 : 11,086	100,000	17,884	17.9%
2	T_2 : 28,968	(in progress)	—	—
3	T_3 : (pending)	—	—	—
Paper (all 3 cycles)		>300,000	>100,000	>80%

4.2 Reproducibility Gaps

Our Cycle 1 class-1 rate (17.9%) is substantially below the paper’s reported >80% across all cycles. This is likely downstream of the classifier gap ($F1 = 0.71$ vs. 0.80), which shifts the decision boundary and affects which generated molecules are classified as effective. We note that:

- The class-1 percentage is expected to increase across cycles as the generative model trains on progressively enriched datasets.
- The 17.9% rate is consistent with our classifier’s lower sensitivity (higher threshold for positive classification).
- The paper’s 80% figure may reflect a different evaluation protocol or classifier configuration that we have not yet identified.

4.3 Workflow Automation

The full pipeline is orchestrated by a single Python script (`run_cycles23.py`) that sequentially: classifies the previous cycle’s output, builds the expanded training set, fine-tunes GPT-2, and generates molecules. Parallel generation shards are launched as independent processes on separate machines, coordinated via SSH.

An AI assistant (OpenClaw/Ollie) manages the infrastructure: deploying conda environments, transferring model weights between machines, monitoring generation progress, and aggregating results. This human-AI collaboration enables rapid iteration on a multi-machine pipeline without manual SSH management.

5 Discussion

5.1 Computational Cost

The primary computational bottleneck is autoregressive generation. GPT-2’s token-by-token sampling means that generating 100,000 valid molecules requires millions of generation attempts, each producing a sequence of up to 100 tokens. Fine-tuning, by contrast, completes in under 2 hours per cycle.

Distributing generation across three machines provides a near-linear speedup (the $2.2\times$ per-machine GPU advantage over CPU, combined with 3-way parallelism, yields $\sim 5.4\times$ total speedup). Further optimization could include:

- Larger batch sizes (limited by GPU memory, though GPT-2 is small enough that this is not a binding constraint)
- Nucleus (top- p) sampling to increase valid molecule rates
- Model distillation to a smaller, faster architecture
- Early termination with reduced targets (e.g., 50K per cycle)

5.2 Reproducibility Considerations

Several aspects of the original paper are underspecified for exact reproduction:

1. The exact SMILES-X hyperparameters (the original SMILES-X paper uses `embed=512`, `LSTM=128`, `dense=128`, but the Fajar et al. paper does not specify which configuration was used).
2. Whether GPT-2 is retrained from scratch each cycle or continued from the previous checkpoint.
3. The exact PubChem query parameters and filtering applied to augmented molecules.
4. The xTB or DFT methodology used for energy gap and dipole calculations (method, basis set, solvent model).

Despite these gaps, the pipeline’s overall architecture is clearly described and the original code repository provides valuable reference implementations.

6 Next Steps

1. Complete Cycles 2 and 3 with parallel generation.
2. Run the full 7-filter pipeline (xTB is installed and ready).
3. Investigate the classifier performance gap by testing with the original SMILES-X hyperparameters (512/128/128).
4. Compare our generated molecule distributions with the paper’s (chemical space coverage, functional group diversity).
5. Assess whether the class-1 percentage increases across cycles as predicted.

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References

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